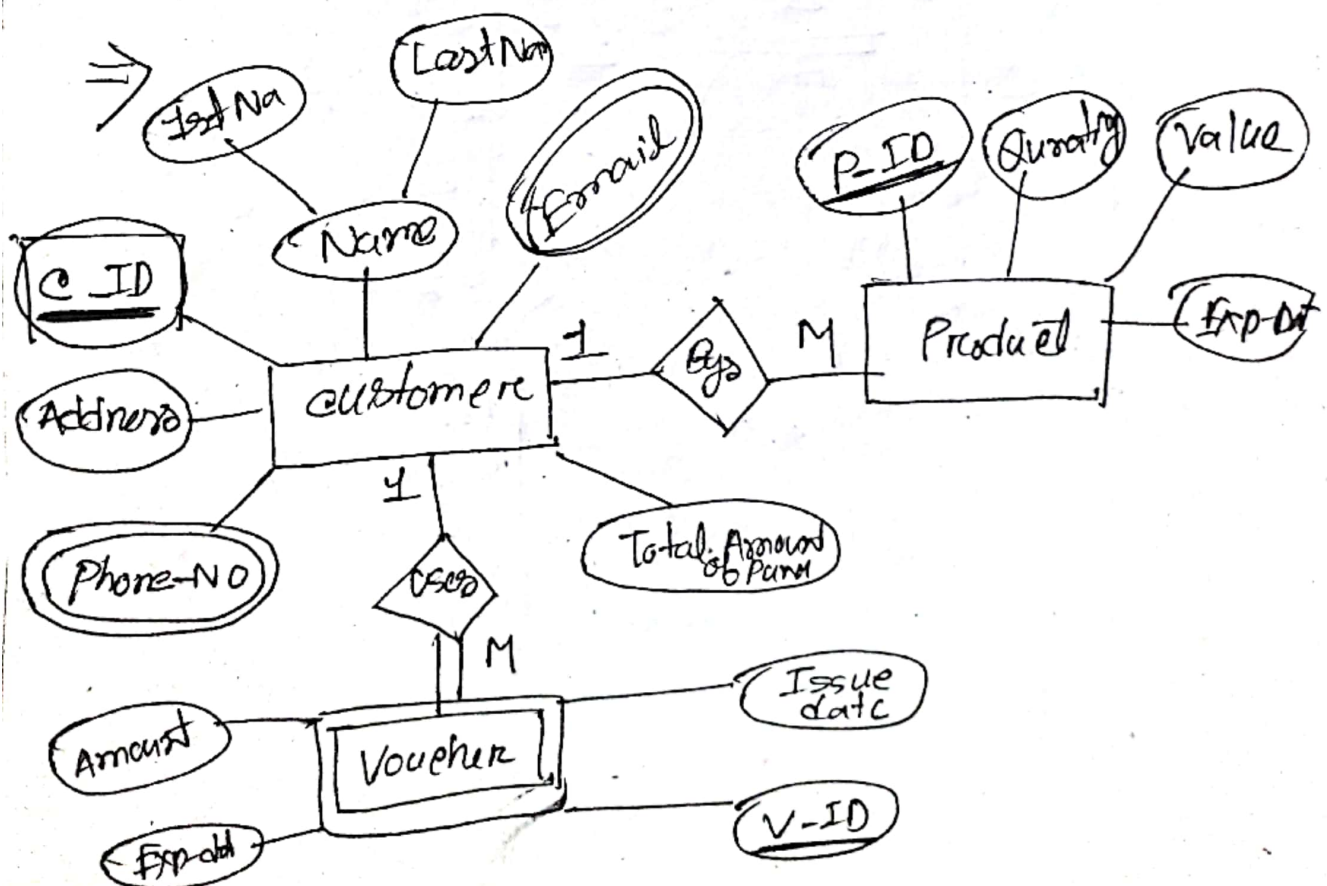


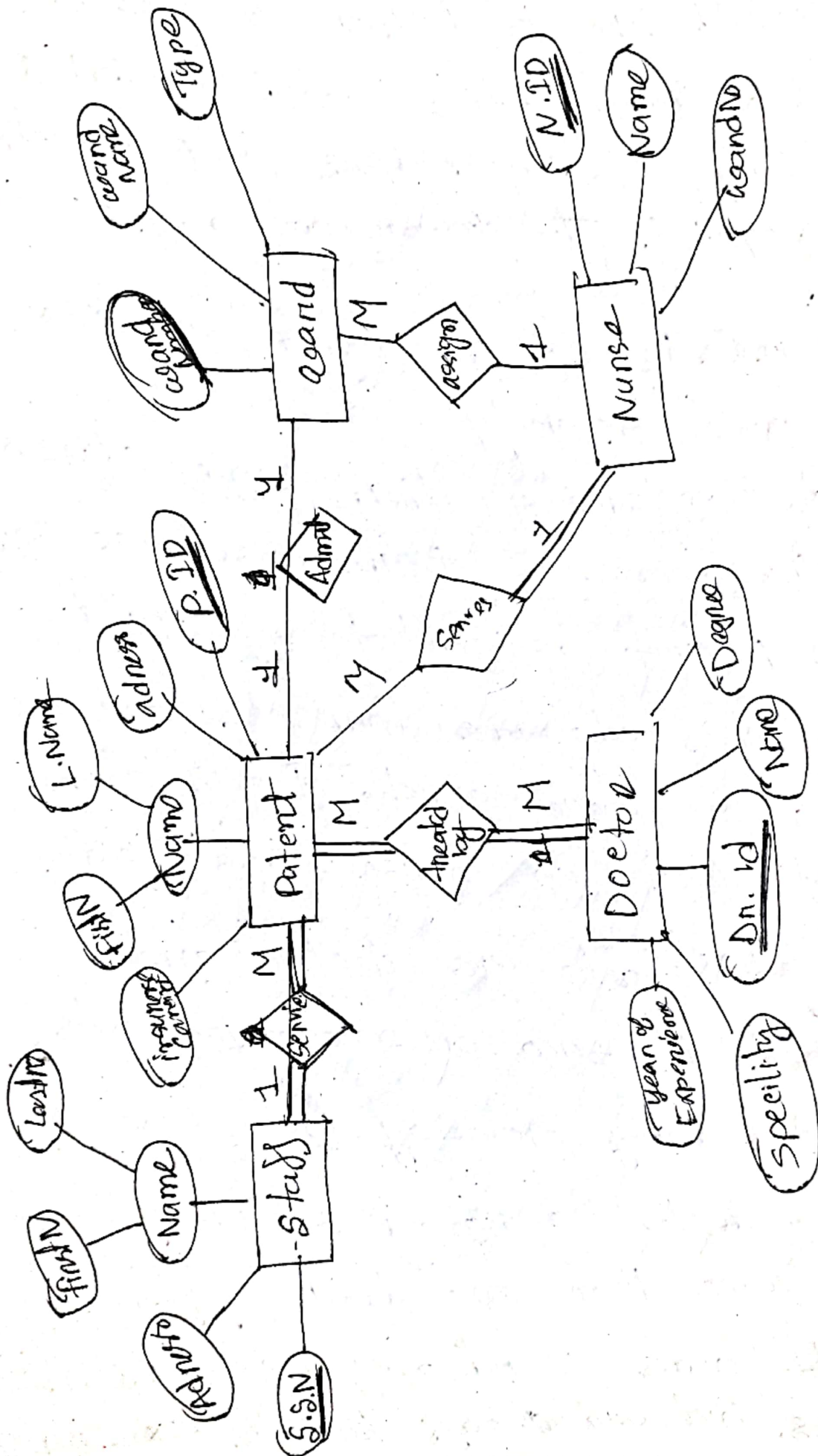
Spring - 18 Jammatul Mauq

20: Construct an F-R diagram for a Grocery Shop with a set of customers and a set of products. Associate with each customer a log of various products and providing each customer with voucher, showing total amount being purchased. Consider the necessary mapping cardinalities and attributes.

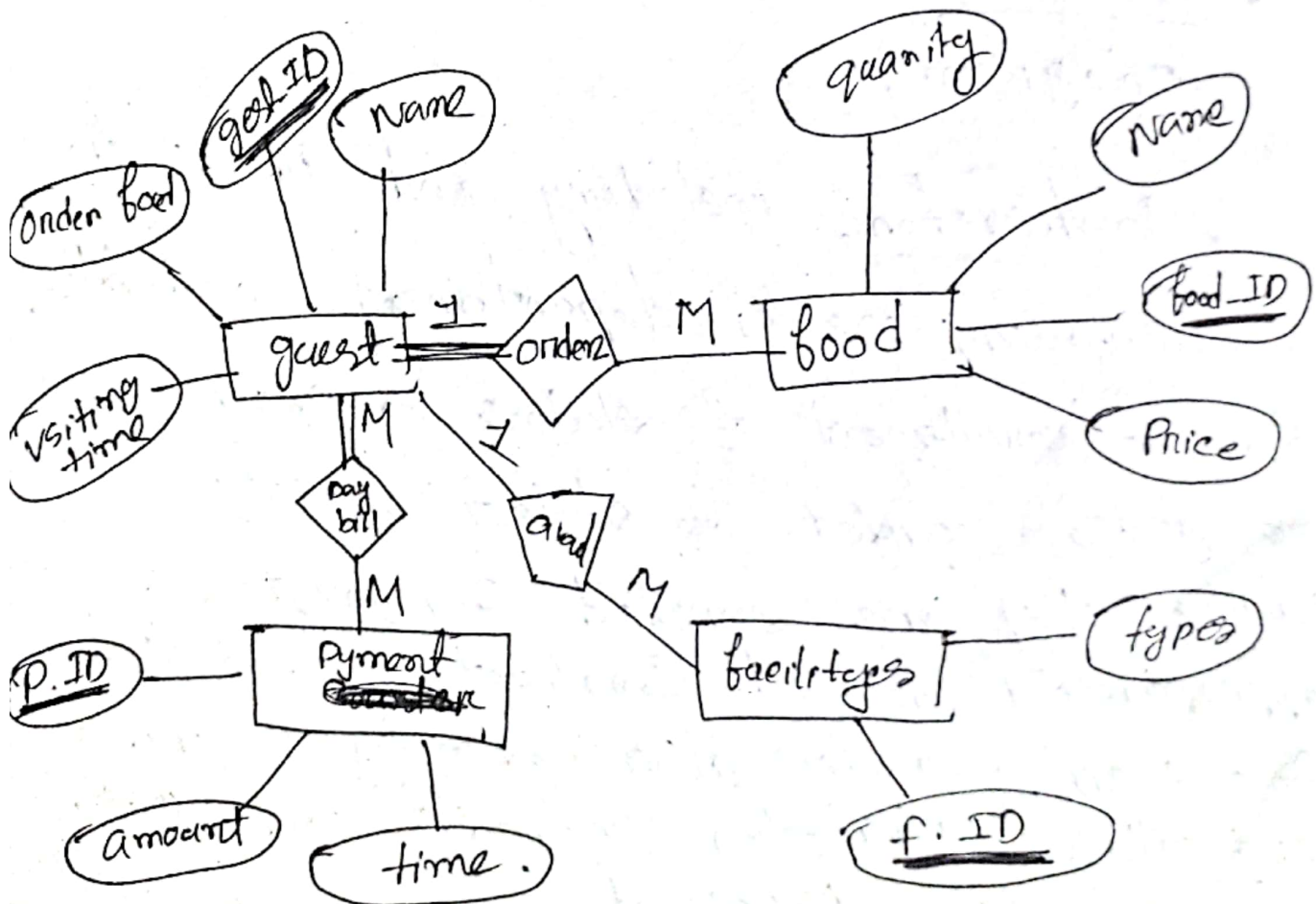


2. (b) we would like to design a database to maintain information about hospital staff including doctors, nurses and patients at the hospital. The information we need includes:

- Staff, including their names, addresses and social security numbers.
- Patients, including their name, address, and the name of their insurance company.
- Each patient assigned to a ward.
- Those staff who are nurses are assigned to zero or more wards. Each ward has at least one nurse assigned.
- The staff who are doctors are assigned to zero or more patients. Patients may or may not have a doctor assigned and they may have more than one doctor. Patients in the same ward may have different doctors, but will always have the same nurse.



Construct an ERD for a restaurant system, where a guest could order any food items available. They could pay their bill through different payment. The guest could also avail many other facilities. Consider the necessary attributes and degrees of relationship.



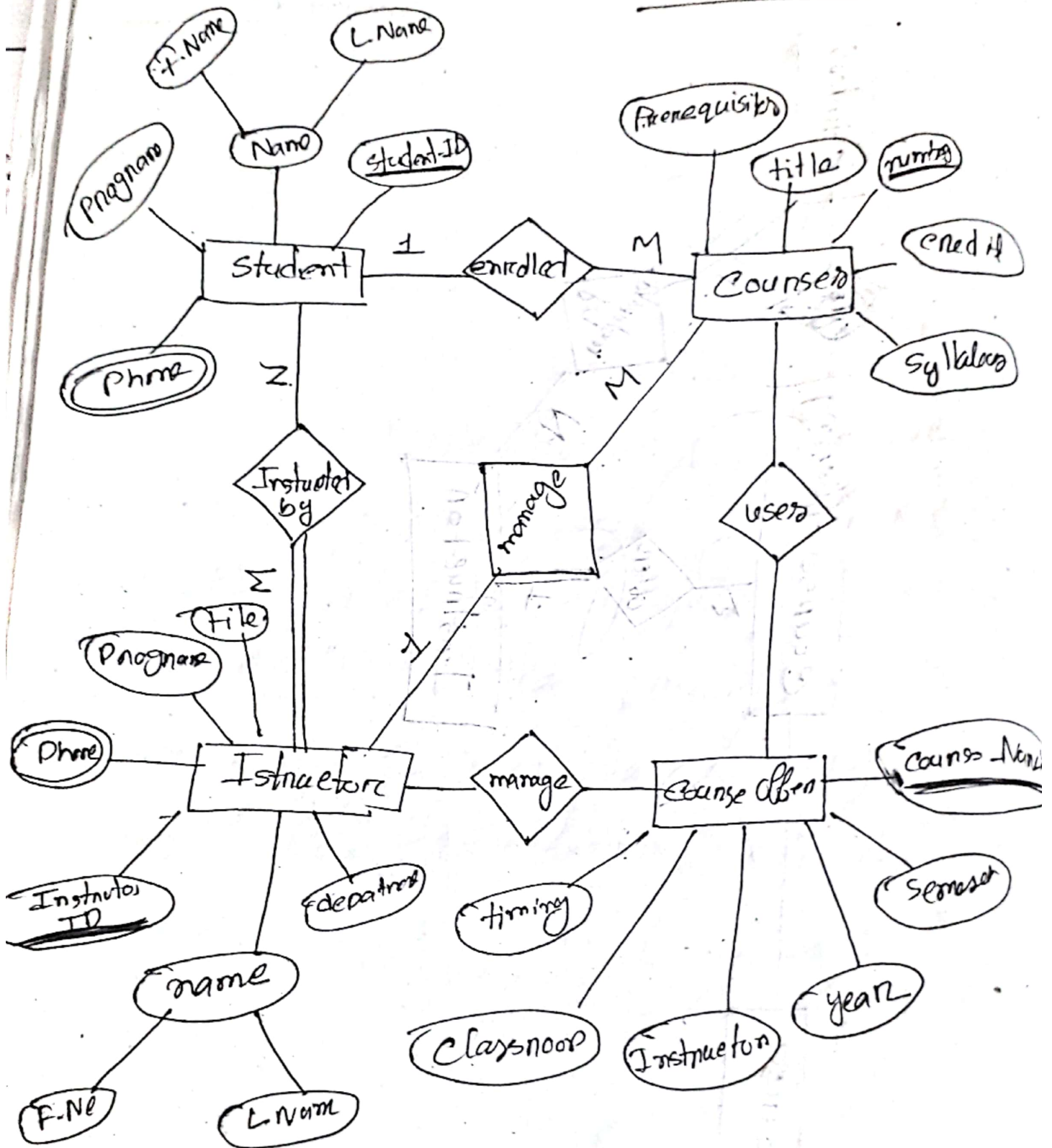
~~2(a) Suppose the overall main~~

2(b) Suppose ACAD maintain data about the following entities:

- Course, including number, title, credits, syllabus, and prerequisites
- Course Offerings, including Course Number, year, semester, Section Number, instructor, timings and classroom.
- Students includes student id, name, program;
- Instructor, including and identification number, name, department and title.

Further, the enrollment of students in course and grades awarded to student in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for ACAD. Consider also necessary mapping and/or participation, constraints and attributes if and where required.

Jannatul Marika



Spring-23

Jannatul Maqsood

⑤ Consider a scenario where a student can register for courses. All the student are enrolled to a specific department which a student is enrolled. A teacher can be a course teacher also an advisor at the same time. Now draw a simple ER diagram from the above scenarios. Consider necessary mapping cardinality, participation constraints and attributes.

